

Jackie Zhang

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EDUCATION

Western University

Master of Engineering in Software Engineering Graduated

London, Canada

Sep 2023 — Aug 2024

- Notable Courses: Machine Learning, Data Management and Applications, Web Application Development

Carleton University

Bachelor of Computer Science (Honors)

Ottawa, Canada

Sep 2019 — May 2023

- Notable Courses: Web Applications Fundamentals, Database Management Systems , Project Management

WORK EXPERIENCE

Full Stack Engineer

Aug 2024 — Present

JoE's Home Renovations & Maintenance

- **LangChain AI Agent Development:** Built production-ready AI agent using LangChain framework with OpenAI GPT-4, implementing AgentExecutor with custom tools (BudgetEstimator, TimelineEstimator, ServiceRecommender) and conversation memory management, achieving 85%+ accuracy for customer renovation queries.
- **End-to-End ML Pipeline:** Contributed to building an end-to-end ML pipeline in Python, covering data preprocessing, feature engineering, model training, and deployment. Implemented and evaluated baseline models using scikit-learn (RandomForest classifier, Gradient Boosting regressor), and exposed predictions through a real-time inference API. The system helped reduce manual estimation effort by approximately 70%.
- **LLM Integration & Fine-tuning:** Integrated OpenAI GPT-4 with LangChain using tool-based agent architecture, fine-tuning prompts for domain-specific renovation knowledge, and implementing context-aware conversation handling for intelligent customer service chatbot.
- **CI/CD & DevOps:** Configured GitHub Actions workflow for AI pipeline with automated testing (pytest), code quality checks (flake8, black, mypy), and integration tests, reducing production bugs by 60%.
- **Data Preprocessing Pipeline:** Designed automated data preprocessing pipeline with text cleaning, feature extraction (description length, word count, budget encoding, date features), missing value handling (multiple strategies), and normalization, processing 1000+ project records efficiently.
- **FastAPI REST API:** Built a RESTful API using FastAPI to serve AI agent and ML model inference, implementing request validation with Pydantic, structured error handling, and basic CORS configuration. Focused on reliability and clean API design for integration with frontend services.
- **Full-Stack AI Integration:** Integrated Python AI backend (LangChain, FastAPI, scikit-learn) with TypeScript/Next.js frontend, creating type-safe API client with fallback mechanisms and seamless user experience.

PROJECT

Cryptocurrency Price Forecasting - LSTM Neural Networks – Project Team Leader

Sept 2023 — Dec 2023

- **Data Engineering:** Built a Python-based ETL pipeline to extract and normalize high-volume on-chain data from Uniswap v3 via The Graph API (GraphQL), with datasets stored and versioned in a cloud-based environment (Azure) for reproducible experiments.
- **Deep Learning:** Designed and evaluated LSTM, GRU, and Transformer architectures using TensorFlow/Keras; optimized the LSTM model to achieve an R^2 of 0.8962, providing high-accuracy price trend forecasts.
- **Feature Analysis:** Conducted noise-correlation experiments on multi-token datasets, proving the impact of cross-asset noise on time-series prediction accuracy.

SKILLS

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript (ES6+), Java, SQL, GraphQL, HTML/CSS.
- **AI/ML Frameworks:** LangChain, OpenAI API, scikit-learn, Pandas, NumPy
- **Frontend:** React, Next.js 14 (App Router), Tailwind CSS
- **Backend & DB:** FastAPI, Next.js API Routes, Prisma ORM, PostgreSQL
- **Cloud & DevOps:** Azure, Google Cloud Platform (GCP), Kubernetes (K8s), Docker, CI/CD (GitLab), Vercel.